Advanced AI In Healthcare for Clinical Decision-Making and Pattern Recognition

For Treatment, Payment, and Operations (TPO)

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Intern Assessment

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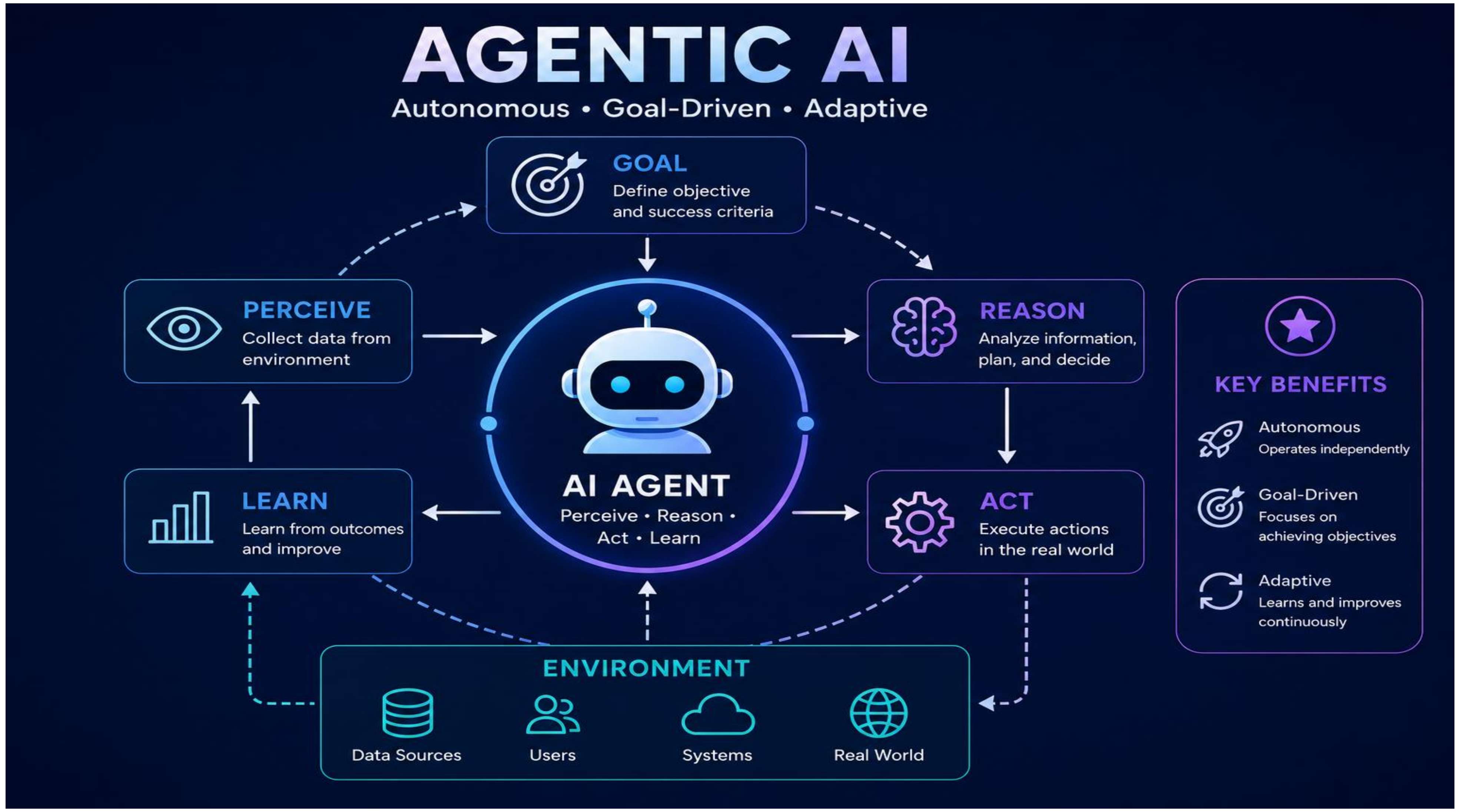
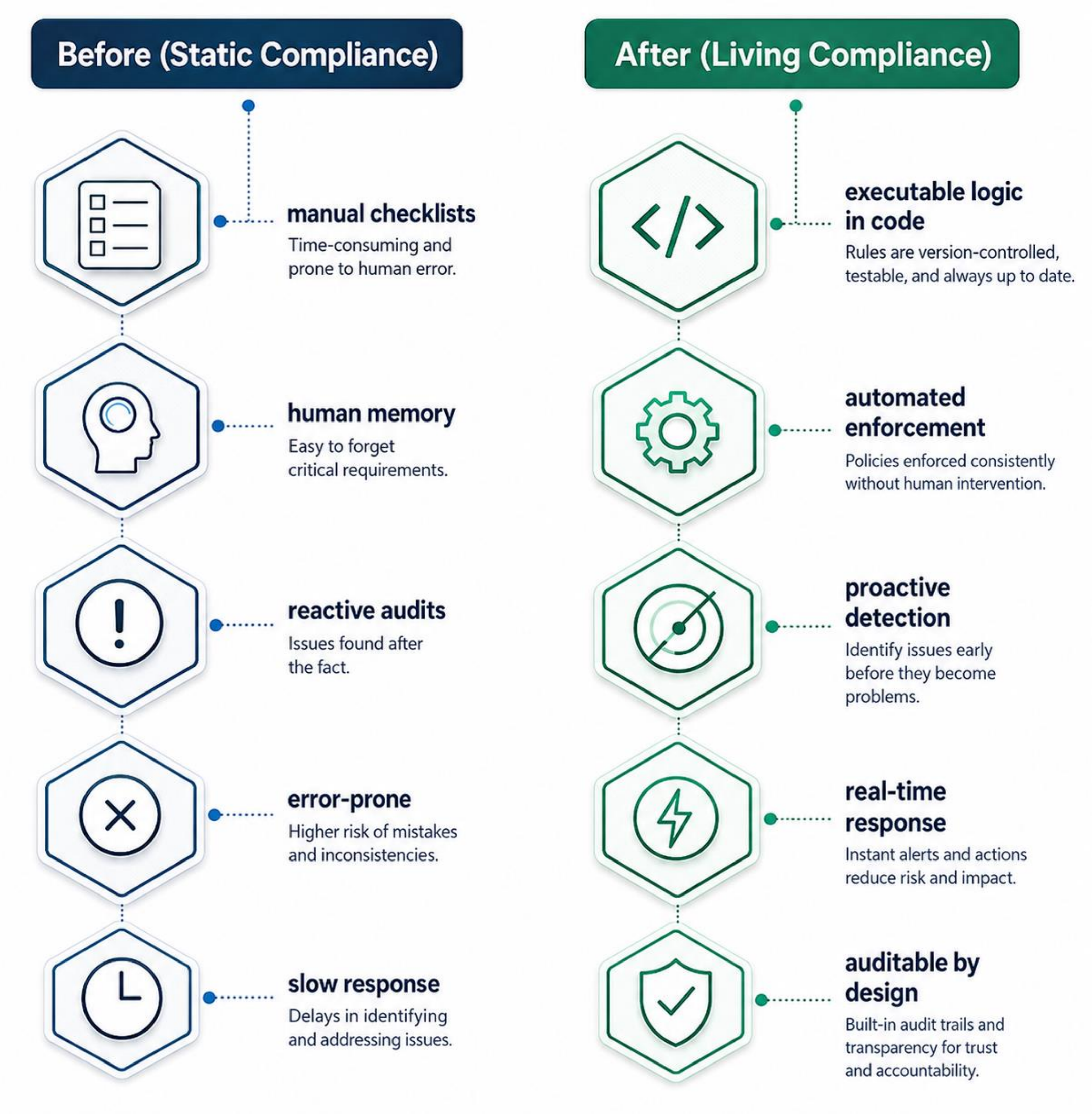
Agentic/Generative AI for Clinical Decision Support in Treatment, Payment, and Operations (TPO)

The healthcare industry is evolving from conversational AI tools (Generative AI) to agentic AI, agents that autonomously analyzes data and coordinates tasks



Agents in TPO workflows:

Agents defines operational objectives, execute end-to-end workflows, and fluidly interact across existing data infrastructures. By it's leverage in deep pattern recognition and chain-of-thought reasoning, an agent processes and breaks down massive, complex data that supports clinical decision making. Agentic AI provides traceable auditable logic in real-time consistency, optimizing data-driven, high-stakes decisions across Treatment, Payment, and Operations (TPO)



Static to Executable Logic

Health Care industries have shifted from static compliance to executable logic where living compliance is embedded, monitored and revised in real time within operational workflows.

Advanced AI has elevated the Healthcare industry by:

1. Enhanced Diagnostics, Workflow Coordination, and Visualization
2. Human manual checklists to Autonomous action lists
3. Improved decisions to achieve desired outcomes
4. Advanced Operational efficiency
5. Guides clinical care and financial outcomes across Treatment, Payment, and Operations (TPO)

TPO: Analytics

Treatment

Continuously Tracks time-series data and uses Deep Learning (DL) diagnostic to classify personalized pathways for care through medical image analysis, pattern recognition, and predictive metrics

Payment

A tracking payment system that automates backend verification to detect claims, payments, and billing errors. It can auto approve or pin-point reasoning of denial

Operations

Continuously tracks statistics and workflow changes, eliminates bottlenecks and directs analytic strategy based on the outcome

Advanced pattern recognition enables organizations to identify operational signals and detect anomalies before risk-based categories escalate

Opportunities

Live Compliance

- Payload validation
- Real-time Tokenization and Privacy
- Continuous Control Monitoring

Accelerated Decisions

- Autonomous Driven Tasks
- Pattern Recognition
- Dynamic self correction

Efficiencies

- Multi-tool orchestration
- Detects payment transactions
- Saves on operational spendings

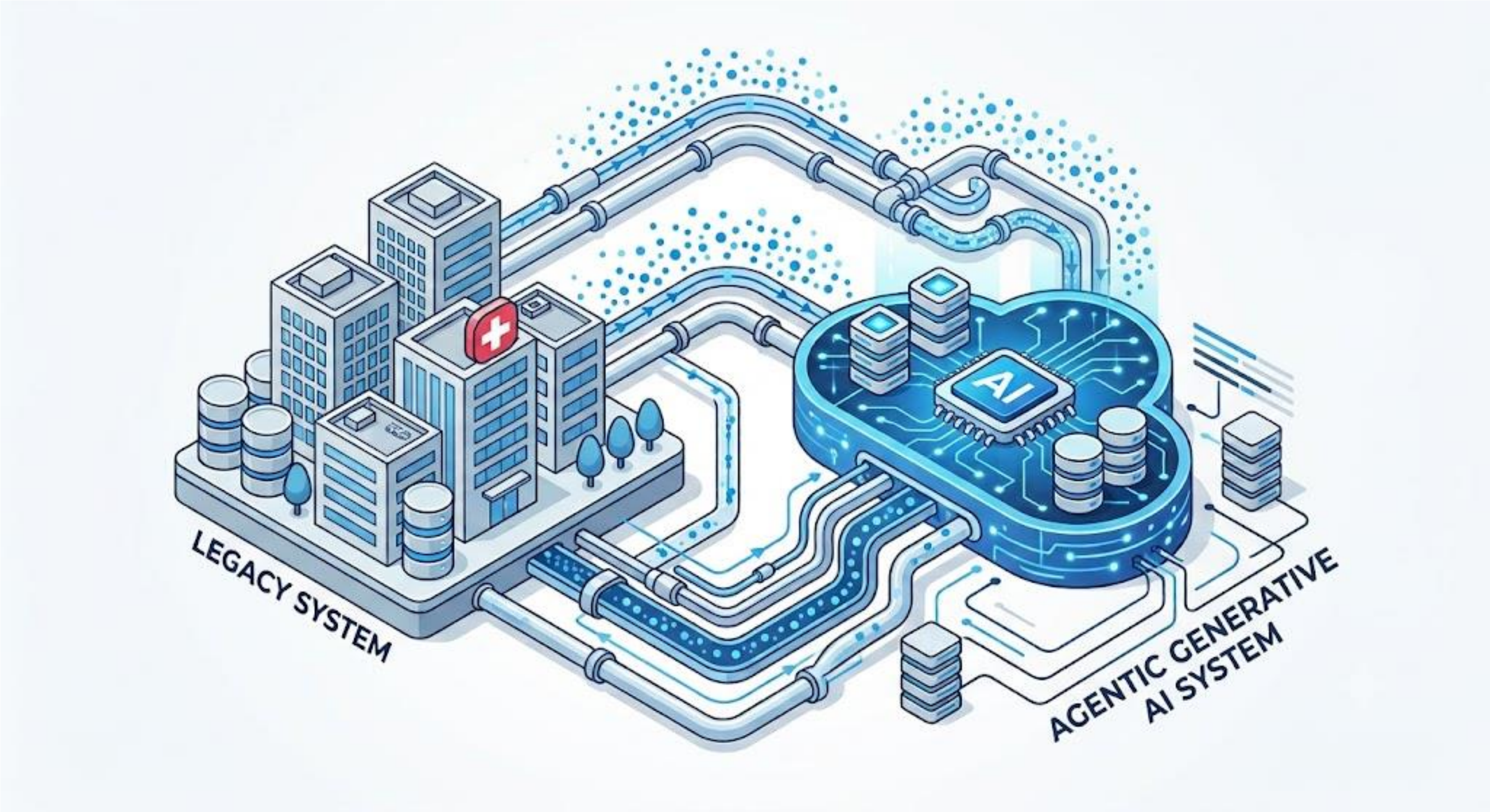
AI Trends In health care

1. Health Care AI Advancement
2. Operation of Autonomous Systems
3. Intelligent Agents
4. Interconnected technology
5. Predictive dynamic models
6. Data Interoperability
7. Time-Series Analysis
8. Dynamic Virtual models
9. Medical intervention / Decision making

+ more

Threat: Operational Delays

Upgrading from legacy systems to AI systems is a transitional risk that is overlooked by health service providers and their patients.



From Legacy to New AI Systems

Building a completely new system alongside legacy infrastructure for AI workflows to be integrated

Protecting Sensitive Data & Rules

Transferring massive amounts of sensitive data and complex rule engines safely

Clinical Operation Deployment Constraints

Experiencing challenges of deployment delays within health care internal systems

Proposal

- Invest in a private generative AI environment for internal company use
- Deploy a secure Agentic AI platform with encryption, tokenization, and key mapping to secure sensitive healthcare data
- Continuously strengthen AI governance, ethical standards, cybersecurity, workforce training, and regulatory compliance as AI technologies and regulations evolve

Objective

Advance Cotiviti's AI strategy by balancing innovation, data security, and responsible AI governance.

Hackathon Proof Of Concept: GitHub Demo

TPO GATEKEEPER: REAL-TIME PAYMENT ENGINE

An automated system that replicates agent driven decision-making to instantly process approvals or denials based on real-time authorization logic.

International Classification of Diseases (ICD)

Prior Authorization Verification

Real-Time Authorization Retrieval

Main Function Deliverables

Living Compliance: The system reads from a version-controlled, hard-coded ruleset that serves as a literal, machine-executable representation of legal contracts.

Generative Autonomy: If a procedure code requires prior authorization and it is missing, the system forces a strict, deterministic rejection—leaving no grey area.

Audit Recorder & Trail: Every decision is permanently recorded for full traceability, allowing for replay. The system generates a log detailing the exact timestamp, the contract rule applied, and the explicit reason for the failure (e.g., missing authorization parameter).

On the readme.md file

Hackathon Proof of Concept

Live Demo

Conclusion

Intelligent agents are becoming increasingly valuable in healthcare operations while improving patient outcomes. The advancement is significant and my interests aligns in the opportunities to improve the quality, efficiency, and accessibility of healthcare services.

Thank you

Thank you for considering me as a potential candidate for the Intern Agentic AI Research SCI role. I look forward to the evaluation process and decision.

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github.com/Ayeitslee/Agentic-Healthcare